Laravel Database Schema

Niki Frozen — Sistem POS

10	10	67	25+
Models	Tabel	Kolom	Relasi

Ringkasan migrasi siap pakai untuk perintah:

```
php artisan make:model NamaModel -mfs
```


Proyek: Sistem POS & Manajemen Stok Niki Frozen

Framework: Laravel (PHP) — Eloquent ORM

Kelompok: FUFU FAFA — Universitas Amikom Yogyakarta

Tahun: 2026

■ Daftar Isi & Ringkasan

No	Model	Tabel	Perintah Artisan	Kolom
1	**Branch**	branches	php artisan make:model Branch -mfs	4
2	**User**	users	php artisan make:model User -mfs	7
3	**Product**	products	php artisan make:model Product -mfs	6
4	**CashierSession**	cashier_sessions	php artisan make:model CashierSession -mfs	8
5	**StockIncoming**	stock_incomings	php artisan make:model StockIncoming -mfs	7
6	**Stock**	stocks	php artisan make:model Stock -mfs	6
7	**Transaction**	transactions	php artisan make:model Transaction -mfs	8
8	**TransactionItem**	transaction_items	php artisan make:model TransactionItem -mfs	9
9	**ExpiryNotification**	expiry_notifications	php artisan make:model ExpiryNotification -mfs	6
10	**Promotion**	promotions	php artisan make:model Promotion -mfs	8

```
// Urutan migrate (ikuti agar foreign key tidak error):
Branch → User → Product → CashierSession → StockIncoming
→ Stock → Transaction → TransactionItem
→ ExpiryNotification → Promotion
```

■■ Catatan Desain Penting

Auto-logout (PB-14)	Gunakan Laravel Sanctum token expiry atau session lifetime di config/session.php — tidak butuh kolom database.
Password (PB-18)	Selalu gunakan Hash::make() bawaan Laravel. Kolom password VARCHAR(255) sudah cukup untuk bcrypt.
Rollback (PB-15)	Implementasi lewat DB::transaction(fn) di service layer. Kolom status='rollback' hanya untuk audit log.
Laporan	Semua laporan (per sesi/harian/bulanan) di-query on-the-fly — tidak perlu tabel terpisah.
Verifikasi Email GM (PB-20)	Gunakan tabel bawaan password_reset_tokens dengan custom flow — tidak butuh tabel baru.
branch_id NULL	Owner & General Manager tidak terikat cabang → branch_id nullable di tabel users.

Model: Branch

Tabel: `branches`

PERINTAH ARTISAN

```
php artisan make:model Branch -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->string('name', 100);
$table->text('address');
$table->timestamps();
```

RELASI

One-to-Many	→ users
One-to-Many	→ products
One-to-Many	→ cashier_sessions
One-to-Many	→ transactions
One-to-Many	→ stocks
One-to-Many	→ stock_incomings

CATATAN SEEDER / FACTORY

■ *Seeder statis saja. Isi 2 data: 'Cabang A' dan 'Cabang B'.*

02 Model: User

Tabel: `users`

PERINTAH ARTISAN

```
php artisan make:model User -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('branch_id')->nullable()->constrained('branches')->onDelete('set null');
$table->string('name', 100);
$table->string('username', 50)->unique();
$table->string('password', 255); // bcrypt via Hash::make()
$table->enum('role', ['owner', 'general_manager', 'admin', 'cashier']);
$table->timestamps();
```

RELASI

Belongs To	← `branches`
One-to-Many	→ `cashier_sessions`

CATATAN SEEDER / FACTORY

■ *Seeder wajib: 1 owner, 1 general_manager, 2 admin (per cabang), 4 cashier (2 per cabang). branch_id NULL untuk owner & general_manager.*

03 Model: Product

Tabel: `products`

PERINTAH ARTISAN

```
php artisan make:model Product -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->string('barcode', 100)->nullable();
$table->string('name', 150);
$table->decimal('price', 10, 2); // harga jual per cabang
$table->timestamps();
```

RELASI

Belongs To	← branches
One-to-Many	→ stock_incomings
One-to-Many	→ stocks
One-to-Many	→ transaction_items
One-to-Many	→ promotions

CATATAN SEEDER / FACTORY

■ *Factory: 20–30 produk frozen food dummy, barcode random, price 15.000–150.000.*

04 Model: CashierSession

Tabel: `cashier_sessions`

PERINTAH ARTISAN

```
php artisan make:model CashierSession -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('user_id')->constrained('users')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->enum('shift', ['morning', 'evening']);
$table->date('date');
$table->timestamp('started_at');
$table->timestamp('ended_at')->nullable(); // null = sesi masih aktif
$table->timestamps();
```

RELASI

Belongs To	← `users`
Belongs To	← `branches`
One-to-Many	→ `transactions`

CATATAN SEEDER / FACTORY

■ *Factory: sesi dummy 30 hari terakhir, kombinasi shift morning/evening per cabang.*

Model: StockIncoming

Tabel: `stock_incomings`

PERINTAH ARTISAN

```
php artisan make:model StockIncoming -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('product_id')->constrained('products')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->foreignId('user_id')->constrained('users')->onDelete('cascade');
$table->integer('quantity');
$table->date('expiry_date'); // wajib diisi — NOT NULL (AC-40)
$table->timestamps();
```

RELASI

Belongs To	← `products`
Belongs To	← `branches`
Belongs To	← `users`
One-to-One	→ `expiry_notifications`

CATATAN SEEDER / FACTORY

■ *Factory: quantity 10–100, expiry_date campuran (sudah lewat / <7 hari / >7 hari) untuk testing notifikasi.*

Model: Stock

Tabel: `stocks`

PERINTAH ARTISAN

```
php artisan make:model Stock -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('product_id')->constrained('products')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->integer('quantity')->default(0);
$table->timestamp('updated_at')->useCurrent()->useCurrentOnUpdate();

$table->unique(['product_id', 'branch_id']); // 1 produk = 1 record per cabang
```

RELASI

Belongs To	← products
Belongs To	← branches

CATATAN SEEDER / FACTORY

■ *Seeder saja: hitung total dari stock_incomings setelah di-seed, masukkan hasilnya. Gunakan updateOrCreate di service layer.*

9 Model: Transaction

Tabel: `transactions`

PERINTAH ARTISAN

```
php artisan make:model Transaction -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('cashier_session_id')->constrained('cashier_sessions')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->string('transaction_code', 50)->unique(); // contoh: TRX-20260607-001
$table->enum('payment_method', ['cash', 'qris']);
$table->decimal('total_price', 12, 2);
$table->enum('status', ['success', 'failed', 'rollback'])->default('success');
$table->timestamps();
```

RELASI

Belongs To	← `cashier_sessions`
Belongs To	← `branches`
One-to-Many	→ `transaction_items`

CATATAN SEEDER / FACTORY

■ *Factory: 200–500 transaksi, 70% cash / 30% qris, status mostly 'success', beberapa 'rollback' untuk uji laporan.*

Model: TransactionItem

Tabel: `transaction_items`

PERINTAH ARTISAN

```
php artisan make:model TransactionItem -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('transaction_id')->constrained('transactions')->onDelete('cascade');
$table->foreignId('product_id')->constrained('products')->onDelete('cascade');
$table->string('product_name', 150); // snapshot nama saat transaksi
$table->decimal('unit_price', 10, 2); // snapshot harga saat transaksi
$table->decimal('discounted_price', 10, 2)->nullable(); // harga setelah promo
$table->integer('quantity');
$table->decimal('subtotal', 12, 2); // (discounted_price ?? unit_price) * quantity
$table->timestamps();
```

RELASI

Belongs To	← `transactions`
Belongs To	← `products`

CATATAN SEEDER / FACTORY

■ *Factory: 1–5 item per transaksi, quantity 1–10, ambil unit_price dari produk terkait.*

09 Model: ExpiryNotification

Tabel: `expiry_notifications`

PERINTAH ARTISAN

```
php artisan make:model ExpiryNotification -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('stock_incoming_id')->constrained('stock_incomings')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->text('message');
$table->boolean('is_read')->default(false);
$table->timestamps();
```

RELASI

Belongs To	← `stock_incomings`
Belongs To	← `branches`

CATATAN SEEDER / FACTORY

■ *Seeder saja: generate dari stock_incomings dengan expiry_date <= 7 hari dari tanggal seed. Jalankan lewat Laravel Scheduler (daily).*

10 Model: Promotion

Tabel: `promotions`

PERINTAH ARTISAN

```
php artisan make:model Promotion -mfs
```

KOLOM MIGRASI

```
$table->id();
$table->foreignId('product_id')->constrained('products')->onDelete('cascade');
$table->foreignId('branch_id')->constrained('branches')->onDelete('cascade');
$table->foreignId('created_by')->constrained('users')->onDelete('cascade');
$table->decimal('discount_percentage', 5, 2); // contoh: 15.00 = 15%
$table->date('valid_until');
$table->boolean('is_active')->default(true);
$table->timestamps();
```

RELASI

Belongs To	← products
Belongs To	← branches
Belongs To	← users (created_by)

CATATAN SEEDER / FACTORY

■ *Factory: produk mendekati kadaluarsa, diskon 10–30%, valid_until dalam 7 hari ke depan.*

■ Semua Perintah Artisan — Copy-Paste Siap Pakai

```
# Jalankan sesuai urutan ini (foreign key dependency):
php artisan make:model Branch -mfs
php artisan make:model User -mfs
php artisan make:model Product -mfs
php artisan make:model CashierSession -mfs
php artisan make:model StockIncoming -mfs
php artisan make:model Stock -mfs
php artisan make:model Transaction -mfs
php artisan make:model TransactionItem -mfs
php artisan make:model ExpiryNotification -mfs
php artisan make:model Promotion -mfs
```

■■ Entity Relationship Diagram (Teks)

```
branches
■■■ users (branch_id → branches.id)
■■■ products (branch_id → branches.id)
■■■ cashier_sessions (branch_id → branches.id)
■■■ stock_incomings (branch_id → branches.id)
■■■ stocks (branch_id → branches.id)
■■■ transactions (branch_id → branches.id)
■■■ expiry_notifs (branch_id → branches.id)
■■■ promotions (branch_id → branches.id)

users
■■■ cashier_sessions (user_id → users.id)

cashier_sessions
■■■ transactions (cashier_session_id → cashier_sessions.id)

products
■■■ stock_incomings (product_id → products.id)
■■■ stocks (product_id → products.id) [UNIQUE: product+branch]
■■■ transaction_items(product_id → products.id)
■■■ promotions (product_id → products.id)

stock_incomings
■■■ expiry_notifications (stock_incoming_id → stock_incomings.id)

transactions
■■■ transaction_items (transaction_id → transactions.id)
```